

Tobi Ayodele

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Education

University of Bristol

Sep 2025 – Jun 2028

BSc (Hons) Computer Science | Second Year

- **Key Modules:** Object-Oriented Programming, Algorithms, Imperative and Functional Programming, Computer Architecture
- Designed a BFS-based Scotland Yard bot with custom heuristics, achieving a win-rate of **60%**. Collaborated using pair programming and Git and presented the algorithm design in an academic viva.

Westcliff High School for Boys

Sep 2018 – Jun 2025

A levels: Computer Science (A), Mathematics (A), Economics (A) Physics (B) EPQ (A)

GCSEs: 12 GCSEs including Computer Science (9) Further Mathematics (8) Mathematics (9) Physics (9)

- Completed EPQ dissertation investigating the advantages and disadvantages of using AI in cybersecurity, including its use in AI-assisted threat detection and the potential for AI-enabled cyberattacks, achieving an **81% mark**.

Experience

Leonardo S.p.A

July 2023 – August 2023

Software Engineering Summer Work Experience

- Collaborated in a **3-person team** to design and build an **autonomous sentry** system according to design specifications simulating a Ministry of Defence security contract.
- Developed a Python-based program using data from the sentry's infrared sensors to enable real-time object detection and autonomous movement.
- Applied Agile methodology to iteratively implement, test, and improve the functionality of the sentry system.
- Pitched the completed system to Leonardo engineers as a security contract bid, explaining design choices, and demonstrating the function of the system within a test track, achieving a **3rd place ranking across Leonardo UK branches**.

Projects

Sudoku Generator and Solver | Java, Python, Git [\[GitHub Repo\]](#)

August 2026 – September 2026

- Developed a Java Sudoku generator, solver, and playable CLI game using recursive backtracking algorithms to generate uniquely solvable puzzles.
- Conducted a **65,000 trial experiment** comparing Random, Greedy/Iterative and Recursive cell-removal strategies and analysed runtime and solution uniqueness using **Python, pandas and Matplotlib**, documenting experimental methodology and findings in a technical [report](#).

Library Management System | Python, SQLite

September 2024 – April 2025

- Developed an object-oriented library management system with a Tkinter GUI supporting **1,000+ members** with role-based functionality for librarians.
- Designed a SQLite relational database for members, books, and loans, implementing search and sorting functionality.
- Conducted client interviews directly with librarians and users to identify problems with current management system and inform requirements
- Implemented login authentication system with secure password hashing, along with testing for input validation and parameterised SQL queries to prevent SQL injection.

Custom Home Lab Machine | Linux, Docker

June 2025 – September 2026

- Built and configured an Ubuntu Server home lab using Docker for self-hosted applications.
- Measured power consumption and analysed estimated annual energy costs, documenting setup and results in a technical [blog post](#).

Skills & Achievements

Languages: Java, Python, C, Haskell, SQL

Technologies & Tools: Linux, Docker, SQLite, Tkinter, Git, GitHub, pandas, Matplotlib

Achievements: **Top 5 finish** in the Senior Mixed category of the national **CyberCenturionX** competition

Leadership: Senior NCO and Platoon Commander, Westcliff Cadet Force. Led field exercises and taught a class of **30+** cadets across the cadet syllabus.